

Exercise Intervention on Young-adult Thalassemia Patients

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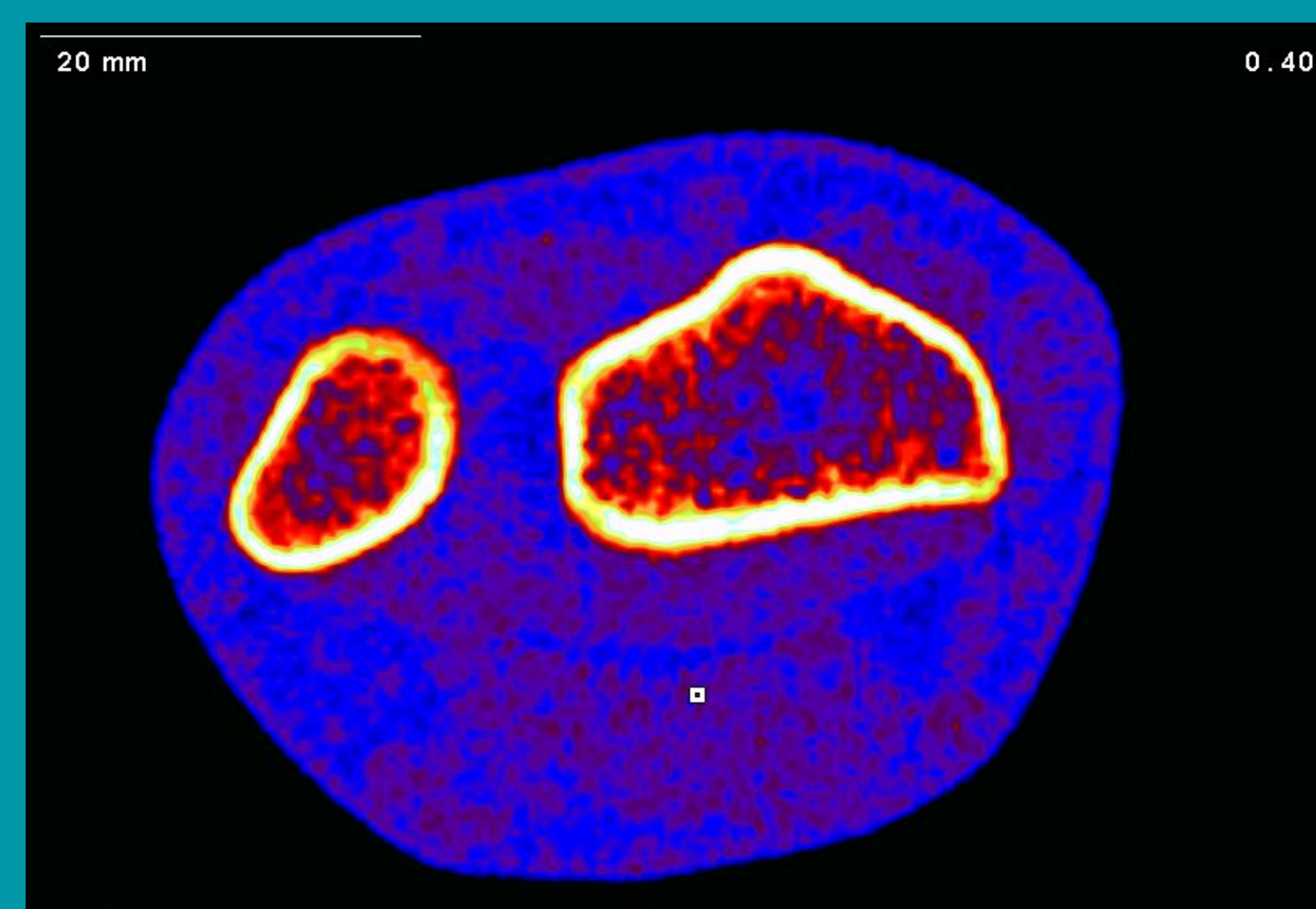
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Intro

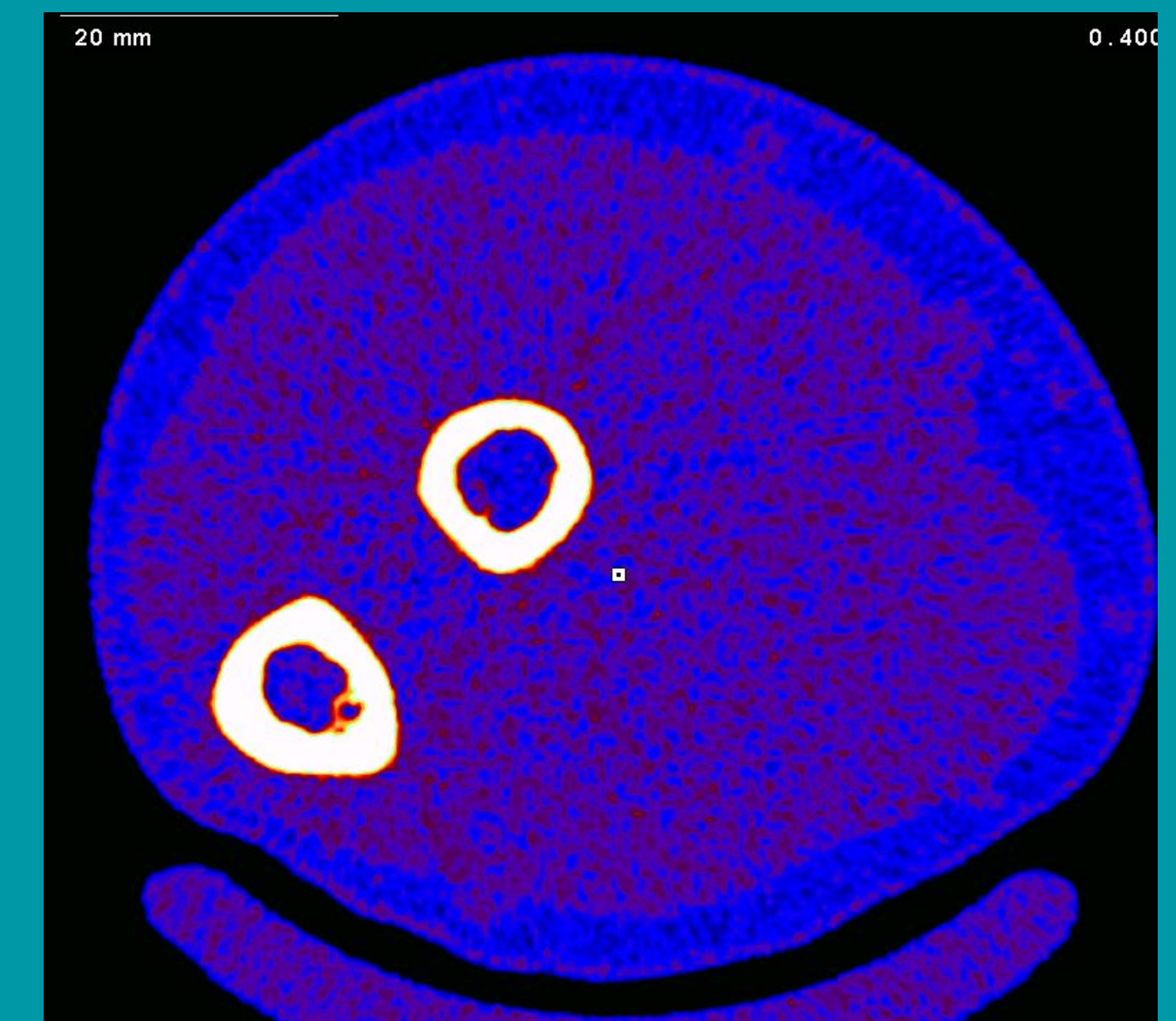
- Thalassemia (Thal) - faulty hemoglobin synthesis.
- 70% of adults with Thal have low bone mass & increased fracture risk.
- Etiology of bone morbidity is multifactorial, but is due in part to inactivity

Methods

- 17 Participants Male and Female (15-35 yrs)
- 3-month home based exercise intervention
- Dietary Intake:
 - 24-hr recall method and analyzed
- Body Composition Assessments:
 - Dual Energy X-ray Absorptiometry (DXA)
 - BOD POD
- Bone Assessments:
 - DXA
 - Peripheral Quantitative Computed Tomography (pQCT)
 - Estimate volumetric BMD at the distal and diaphyseal radius and tibia, bone geometry and bone strength
- Functional Assessments:
 - Explore muscle function and lower limb power
 - Maximal hand grip strength
 - Hand dynamometer
 - Predictor of cortical bone strength
 - The Vertical Jump test
 - Used to estimate lower body power - Vertec
 - Noninvasive predictor of tibial bone strength parameters
 - Sit to Stand Assessment (STS)
 - Estimate lower body power - GymAware technology
 - Accessible, cost-effective method for evaluating lower body power
 - The standardized 6 min walk test
 - Marker of endurance

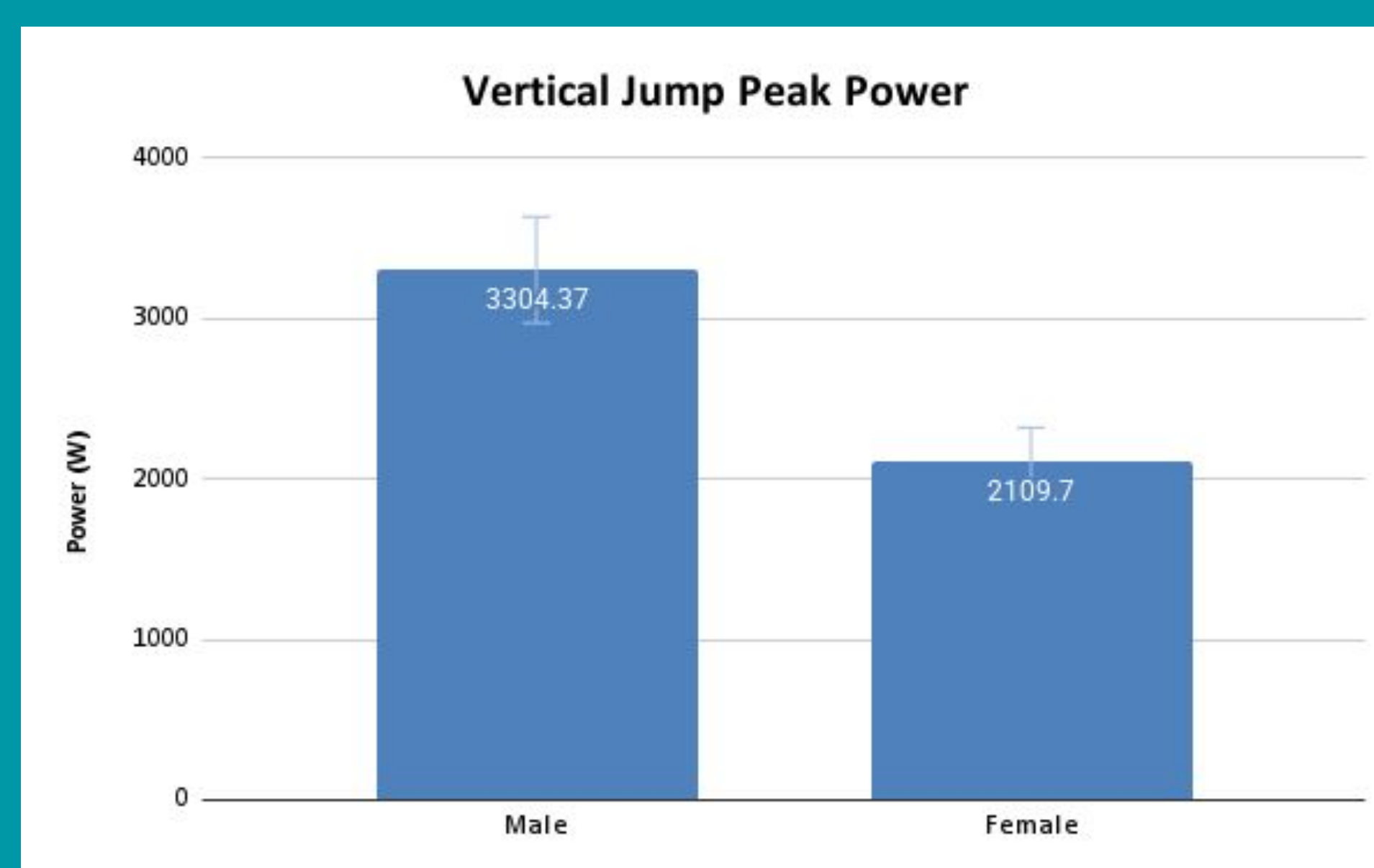


4% site of Ulna and Radius

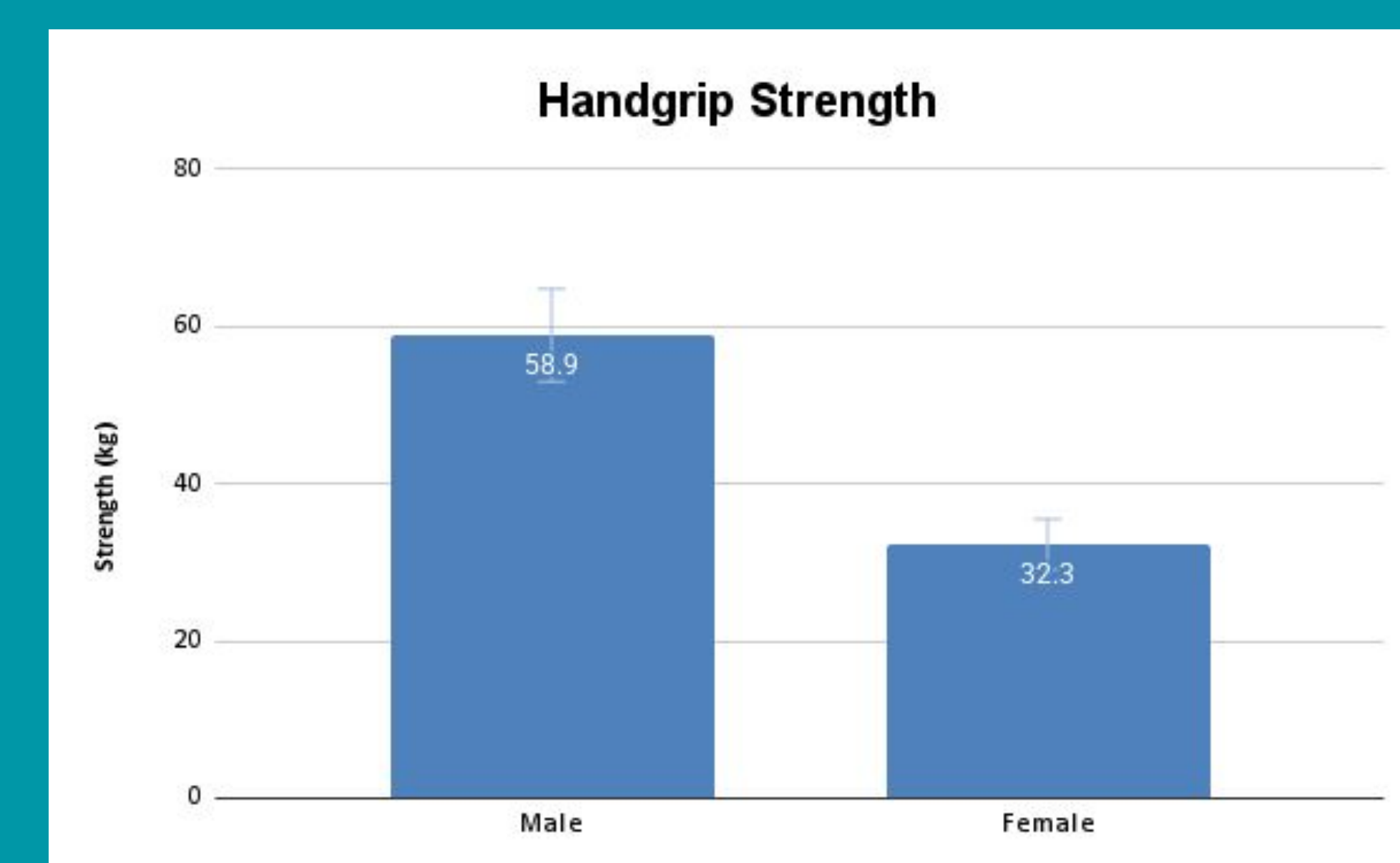


40% site of Ulna and Radius

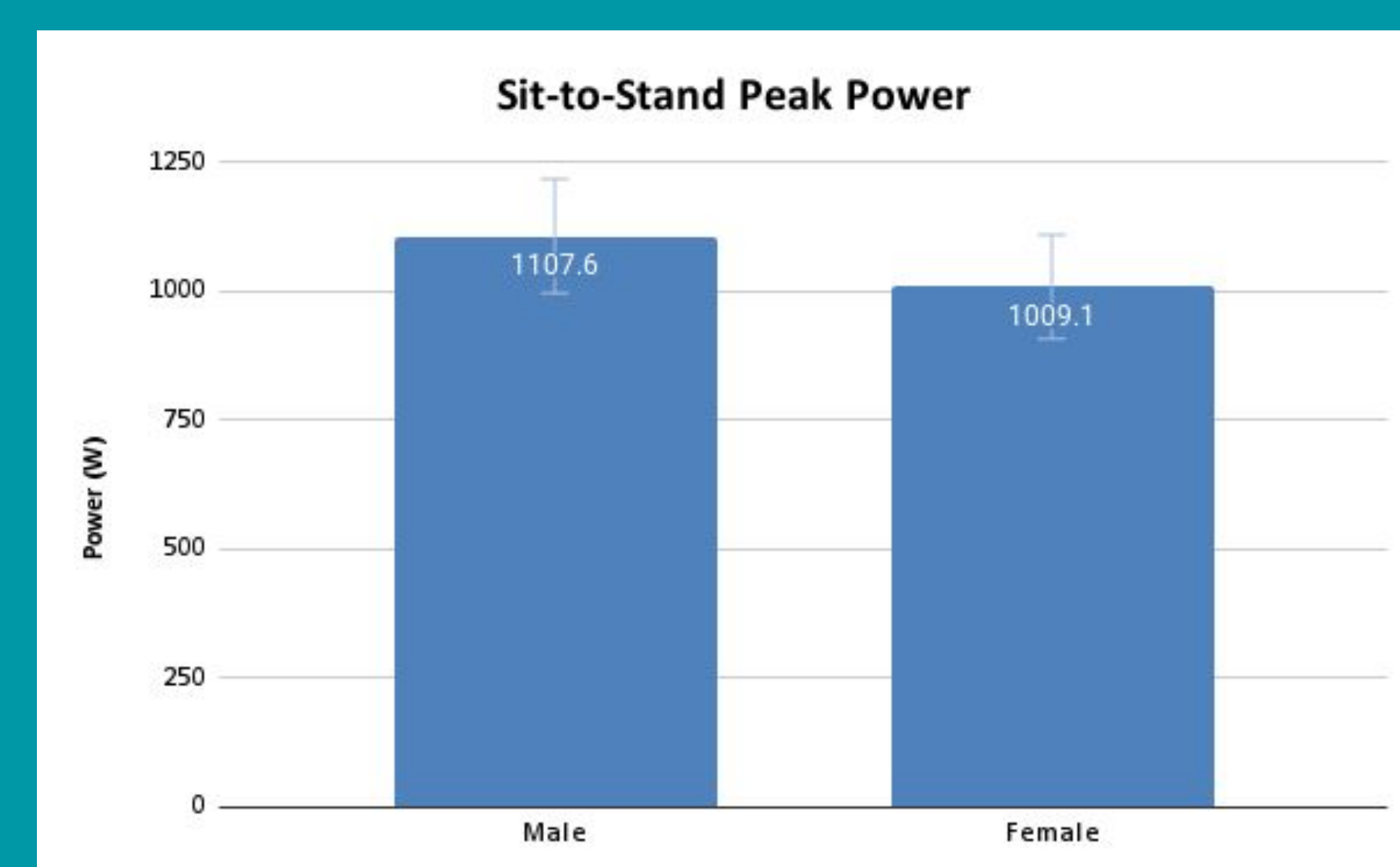
Results



Male and Female VJ Peak Power: Thal females were 65% of age match and males were 73%



Male and female GS: Thal participants had same power as pop of older pop of over 65



Sit-to-Stand: Males were 62-64% of age match and females were 55-57%

Take Home

Muscle function is lower in Thal patients, which means that it is probable that bone strength will be lower too

Next Steps

- Determine if exercise results in increased muscle function and bone strength
- Continue with exercise program and monitoring daily activity for each participant